Scope

1. Definition of the CSP Framework and Physical State Variables.
2. Definition of the Network and Interaction and Their Roles in the CSP Framework.
3. Conservation Properties and Continuum Consistency.

-Discrete level: mass, momentum, energy conservation.

-continuum limit: continuum consistency of momentum equation. (continuity equation needed.)

4. Derivation of the CSP Equations of Motion from the Euler-Lagrange Equation.
5. Theoretical Comparison with SPH.

Introduction

"The objective of this work is not to introduce a new mathematical theory, but to establish a new physics-based computational framework for continuum mechanics. Mathematics is employed solely as a rigorous language to describe the proposed physical concepts."

Assumption, Axioms, and Definitions

Assumption: Continuum Hypothesis

-The fluid is treated as a continuum, such that macroscopic physical fields are well-defined throughout the computational domain.

Axiom

Axiom 1. Geometric Space

-Fluid exists in a geometric space, denoted by Ω. This space provides a geometric structure in which the position of every computational state point is well-defined, while remaining independent of any particular coordinate system.

Axiom 2. Time

-Time is a continuous parameter that governs the evolution of the physical states of computational state points, thereby defining the order of physical events.

Definitions

1. Computational State Point

-A computational state point (CSP) is a computational entity representing the local physical state of a continuum at a given location. The CSPs are neither physical particles nor finite material elements of the continuum, but they exhibit particle-like characteristics in their individual evolution, while the network provides the mechanism for describing continuum behavior from interactions among CSPs. Each CSP possesses an invariant mass, while no intrinsic volume is assigned to it.

2. Virtual Volume

-A virtual volume is a computational volume assigned to a Computational State Point for defining volume-dependent physical quantities. It does not represent the physical volume occupied by the material, but serves solely as a computational construct.

3. Physical State

-The physical state of a Computational State Point is the collection of physical quantities describing its instantaneous mechanical and thermodynamic condition. The state variables may be classified into primitive and derived variables. Primitive variables are directly assigned to a CSP, whereas derived variables are evaluated from primitive variables together with the associated virtual volume.

4. Interaction

-The energy coupling between CSPs induced by their Network-dependent physical states.

5. Network

-A computational structure that decides the virtual volume of each CSP. It quantifies the geometric relations among CSPs.

Principles

Fundamental Principles

1. Independent Interaction Principle

Proof using 1D inverse-distance Network

- Mass of each CSP is same as m.
- Total volume V_{total} is constant.
- Enough amount of CSPs are exist, so the SUM can be converted into integration.
- Boundary conditions are ignored or assumed as periodic boundary.

1. Network

1D, 3 CSPs (i-1, i, i+1).

$$N_i = \left(\frac{1}{x_i - x_{i-1}} + \frac{1}{x_{i+1} - x_i} \right)^{-1}$$

$$V_i = V_{total} \frac{N_i}{\sum_k N_k}$$

2. Under equal distance condition

If $x_i - x_{i-1} = x_{i+1} - x_i = \Delta x$,

$$N_i = \left(\frac{1}{\Delta x} + \frac{1}{\Delta x} \right)^{-1} = \frac{\Delta x}{2}$$

If there are n CSPs,

$$V_{total} = n \Delta x$$

$$\sum_k N_k = n \frac{\Delta x}{2} = \frac{V_{total}}{2}$$

$$V_i = V_{total} \frac{\Delta x/2}{V_{total}/2} = \Delta x$$

3. multiple CSPs

$$x_i - x_{i-1} \approx x_{i+1} - x_i \approx \Delta x$$

$$N_i \approx \frac{\Delta x}{2}$$

$$\sum_k \Delta x \to V_{total},$$

$$\sum_k N_k \approx \frac{1}{2} \sum_k \Delta x \to \frac{V_{total}}{2}$$

$$V_i \to V_{total} \frac{\Delta x/2}{V_{total}/2} = \Delta x$$

4. Density

$$\rho_i = \frac{m_i}{V_i}, \quad V_i \to \Delta x$$

$$\rho_i \to \frac{m_i}{\Delta x}$$

5. Continuity

$$m_i = \rho_i V_i = constant$$

$$\frac{d}{dt}(\rho_i V_i) = 0$$

$$V_i \frac{d\rho_i}{dt} + \rho_i \frac{dV_i}{dt} = 0$$

$$\frac{d\rho_i}{dt} + \frac{\rho_i}{V_i}\frac{dV_i}{dt} = 0$$

$$\frac{dV_i}{dt} \to u_{i+1} - u_i$$

$$\frac{1}{V_i}\frac{dV_i}{dt} \to \frac{u_{i+1} - u_i}{\Delta x}$$

$$\lim_{\Delta x \to 0} \frac{u_{i+1} - u_i}{\Delta x} = \frac{\partial u}{\partial x}$$

$$\frac{d\rho_i}{dt} + \rho\frac{\partial u}{\partial x} = \frac{\partial \rho}{\partial t} + \frac{\partial \rho}{\partial x}\frac{dx}{dt} + \rho\frac{\partial u}{\partial x} = \frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x}$$

6. Pressure force

$$U = \sum_k m_k e(\rho_k), \quad F_i = -\frac{\partial U}{\partial x_i}, \quad \frac{\partial U}{\partial V_k} = -P_k$$

$$F_i = \sum_k P_k \frac{\partial V_k}{\partial x_i}$$

Under 1D, $A = 1$

$$P = \frac{F}{A} = F, \quad F_i = P_{left} - P_{right} = -\left(P_{i+1/2} - P_{i-1/2}\right)$$

$$\frac{F_i}{V_i} \to -\frac{P_{i+1/2} - P_{i-1/2}}{\Delta x} \to -\frac{\partial P}{\partial x}$$

$$\frac{m_i}{V_i}\frac{du_i}{dt} = \rho_i \frac{du_i}{dt} = \frac{F_i}{V_i}$$

$$\rho\frac{Du}{Dt} = -\frac{\partial P}{\partial x}$$

7. Coliision

$$r_{ij} = \left| x_i - x_j \right| \to 0$$

$$\frac{1}{r_{ij}} \to +\infty$$

$$N_i \to 0$$

when $\sum_k N_k > 0$,

$$V_i = V_{total} \frac{N_i}{\sum_k N_k} \to 0$$

$$\rho_i = \frac{m_i}{V_i} \to +\infty$$

If $\rho \to \infty \Rightarrow e(\rho) \to \infty$,

$$U_i = m_i e(\rho_i) \to +\infty$$

If a conservative system has finite total energy, collision is impossible.

Purpose: $\frac{F_i}{V_i} \rightarrow -\nabla P$

Assumption.

- Mass of each CSP is same as m.
- Total volume V_{total} is constant.
- Kernel W_h is symmetrical.

$$W_h(r) = W_h(-r)$$

- Enough amount of CSPs are exist, so the SUM can be converted into integration.
- Boundary conditions are ignored or assumed as periodic boundary.

1. Network.

$$N_i = \frac{1}{n_i},\ n_i = \sum_j W_h(x_i - x_j).$$

(n_i is the value that represents the degree of the CSP crowding nearby the CSP_i.)

2. Definition of Virtual Volume.

$$V_i = V_{toal}\frac{N_i}{\sum_k N_k}$$

For the convenience, let $S = \sum_k N_k$ and $\alpha = \frac{V_{total}}{S}$.

Then, $V_i = \alpha N_i$.

($\sum_i V_i = \alpha \sum_i N_i = \frac{V_{total}}{S} S = V_{total}$.)

3. Definition of Density.

$$\rho_i = \frac{m}{V_i}.$$

4. Definition of Internal Energy.

Let the specific internal energy $e = e(\rho)$.

$U_i = me(\rho_i)$,

$U = \sum_k me(\rho_k)$.

($P(\rho) = \rho^2 \frac{de}{d\rho}$ is a thermodynamic relation.)

5. Differentiate U_i with respect to the V_i.

$$\frac{\partial \rho_i}{\partial V_i} = \frac{\partial}{\partial V_i}\left(\frac{m}{V_i}\right) = -\frac{m}{V_i^2}.$$

$$\frac{\partial U_i}{\partial V_i} = m\frac{\partial e}{\partial V_i} = m\frac{de}{d\rho_i}\frac{\partial \rho_i}{\partial V_i} = -m\frac{de}{d\rho}\frac{m}{V_i^2}.$$

Since $\frac{m}{V_i} = \rho_i$,

$$\frac{m^2}{V_i^2} = \rho_i^2, \ \frac{\partial U_i}{\partial V_i} = -\rho_i^2\frac{de}{d\rho_i}.$$

Since $P_i = \rho_i^2\frac{de}{d\rho_i}$,

$$\frac{\partial U_i}{\partial V_i} = -P_i, \ P = -\frac{\partial U}{\partial V}.$$

6. Euler-Lagrange Equation.

$$T = \sum_i \frac{1}{2} m_i |\dot{x}_i|^2, \ L = T - U.$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_i}\right) - \frac{\partial L}{\partial x_i} = 0.$$

$$\frac{d}{dt}\left(m\dot{x}_i\right) + \frac{\partial U}{\partial x_i} = 0, \ m\ddot{x}_i = -\frac{\partial U}{\partial x_i},$$

$$ma_i = -\nabla_i U.$$

7. Definition of the Force.

Let $F_i = -\nabla_{x_i} U$.

$$\nabla_{x_i} U = \sum_k \frac{\partial U}{\partial V_k}\nabla_{x_i} V_k.$$

Since $\frac{\partial U}{\partial V_k} = -P_k$,

$$\nabla_{x_i} U = -\sum_k P_k \nabla_{x_i} V_k, \ F_i = \sum_k P_k \nabla_{x_i} V_k.$$

8. Differentiate of V_k.

$$V_k = V_{total}\frac{N_k}{S} = \alpha N_k$$

$$\nabla_{x_i} V_k = V_{total}\left(\frac{S\nabla_{x_i} N_k - N_k \nabla_{x_i} S}{S^2}\right) = \alpha\left[\nabla_{x_i} N_k - \frac{N_k}{S}\nabla_{x_i} S\right].$$

$$F_i = \alpha\sum_k P_k\left[\nabla_i N_k - \frac{N_k}{S}\nabla_i S\right] = \alpha\left[\sum_k P_k \nabla_i N_k - \frac{\sum_k P_k N_k}{S}\nabla_i S\right].$$

Let $\overline{P}_N = \dfrac{\sum_k P_k N_k}{S}$.

Since $S = \sum_k N_k$,

$$F_i = \alpha\sum_k \left(P_k - \overline{P}_N\right)\nabla_i N_k.$$

9. Differentiate N_i.

Since $N_i = \dfrac{1}{n_i}$, $n_i = \sum_j W_{ij}$,

$$\nabla_i N_k = -\frac{1}{n_k^2}\nabla_i n_k = -N_k^2 \nabla_i n_k.$$

$$\nabla_i n_k = \sum_j \nabla_i W_{kj}.$$

when $k \neq i$,

$$\sum_j \nabla_i W_{kj} = \nabla_i W_{ki} = \nabla_i W_h\left(x_k - x_i\right).$$

$$\nabla_i N_k = -N_k^2 \nabla_i W_{ik}.$$

10. Inserting to Force.

$$F_i = \alpha\sum_k \left(P_k - \overline{P}_N\right)\nabla_i N_k$$

dividing the force equation to the terms, $k = i$ and $k \neq i$,

$$F_i = -\alpha\left(P_i - \overline{P}_N\right)N_i^2 \sum_j \nabla_i W_{ij} - \alpha\sum_{j\neq i}\left(P_j - \overline{P}_N\right)N_j^2 \nabla_i W_{ij},$$

$$F_i = -\alpha\sum_{j\neq i}\left[\left(P_i - \overline{P}_N\right)N_i^2 + \left(P_j - \overline{P}_N\right)N_j^2\right]\nabla_i W_{ij}.$$

11. Continuum Limit.

Let the CSP number density of space is $q(x)$.

$$n_i = \sum_j W_h\left(x_i - x_j\right),\ n_i \to q(x_i).$$

$N_i \to \frac{1}{q_i}.$

12. Normalization α.

$S = \sum_i N_i.$

at the continuum limit, $\sum_i f_i \to \int_\Omega f(x) q(x) dV.$

Therefore, $S \to \int_\Omega \frac{1}{q(x)} q(x) dV = \int_\Omega dV = |\Omega|.$

When $V_{total} = |\Omega|$,

$\alpha \to 1, \; V_i \to \frac{1}{q_i}.$

Since $\rho = mq, \; V_i \to \frac{m}{\rho_i}.$

13. Convert Force Sum to Integration.

Since $\left(P_i - \overline{P}\right) N_i^2 \to \frac{P - \overline{P}}{q^2},$

Let $A(x) = \frac{P(x) - \overline{P}}{q(x)^2}.$

at the continuum, $F(x) = -\int [A(x) + A(y)] \nabla_x Wh(x-y) q(y) dy.$

14. First Part of the Integration.

$-A(x) \int \nabla_x W_h(x-y) q(y) dy.$

According to the Kernel Convolution,

$\int W_h(x-y) q(y) dy \to q(x).$

$\int \nabla_x W_h(x-y) q(y) dy \to \nabla q.$

Therefore, the first term is $-A \nabla q = -\frac{P - \overline{P}}{q^2} \nabla q.$

15. Second Part of the Integration.

$-\int A(y) q(y) \nabla_x W_h(x-y) dy$

$Aq = \dfrac{P - \overline{P}}{q}$, let $Aq = g(y)$.

$$-\int g(y)\nabla_x W_h(x-y)dy = -\nabla_x \int g(y)\, W_h(x-y)dy.$$

According to the Kernel Convolution,

$$-\int g(y)\nabla_x W_h(x-y)dy = -\nabla_x (W_h * g)(x)$$

When $h \to 0$,

$$W_h * g \to g,\ -\nabla\left(\frac{P-\overline{P}}{q}\right).$$

16. Combine.

$$F \to -\frac{P-\overline{P}}{q^2}\nabla q - \nabla\left(\frac{P-\overline{P}}{q}\right).$$

Since $\nabla \overline{P} = 0$,

$$\nabla\left(\frac{P-\overline{P}}{q}\right) = \frac{1}{q}\nabla P - \frac{P-\overline{P}}{q^2}\nabla q.$$

$$F = \frac{P-\overline{P}}{q^2}\nabla q - \left[\frac{1}{q}\nabla P - \frac{P-\overline{P}}{q^2}\nabla q\right] = -\frac{1}{q}\nabla P.$$

17. Arrangement.

Since $V = \dfrac{1}{q}$,

$$F = -V\nabla P,\ \frac{F}{V} = -\nabla P.$$

Since $m = \rho V$ and $ma = F$,

$\rho Va = -V\nabla P,\ \rho a = -\nabla P.$

If CSP moves like Largrangian material point,

$$a = \frac{Du}{Dt},$$

$$\rho\frac{Du}{Dt} = -\nabla P.$$